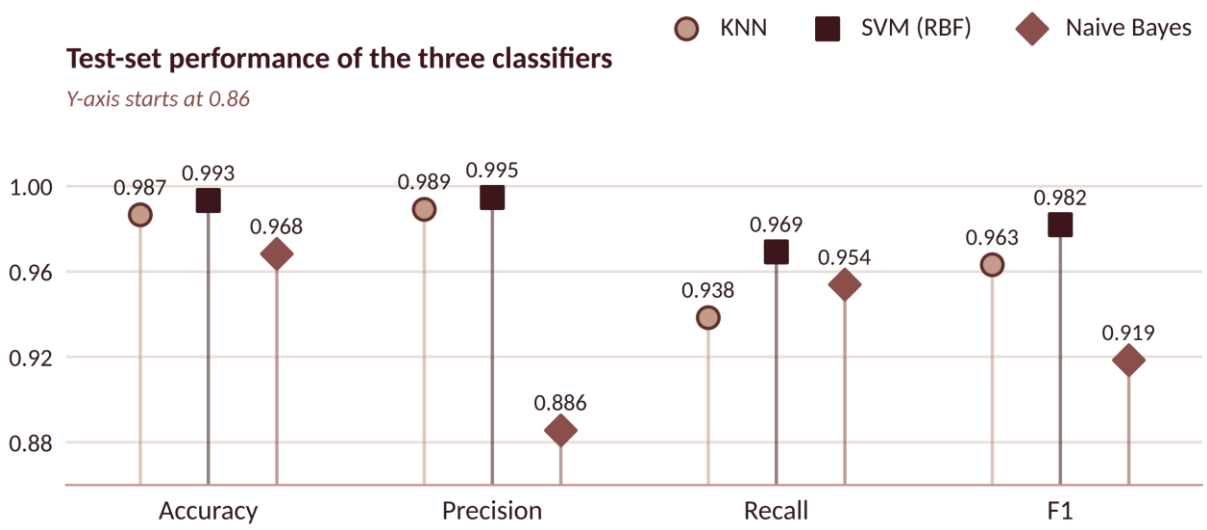
PART G · FINAL REPORT

Final Analysis & Reporting



Setup (from the notebook). 5,200 messages; 9 behavioural features (num_special_chars and both keyword scores were dropped in Part B); stratified 80/20 split into 4,160 training and 1,040 test messages (195 spam); median imputation and standard scaling fitted on the training set; KNN with k = 5, SVC with RBF and linear kernels, GaussianNB. Five-fold cross-validation on the training set is added to confirm the ranking is not a single-split accident.

1 Strengths and weaknesses of each classifier

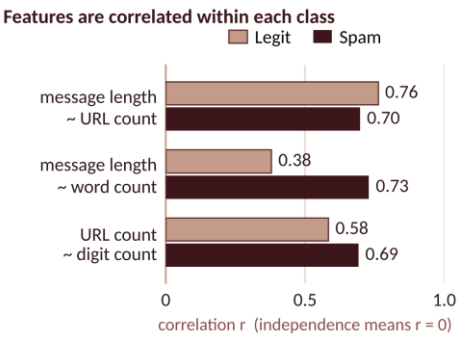
Model	Accuracy	Precision	Recall	F1	FP	FN	Fit (ms)	Predict (ms)
KNN	0.987	0.989	0.938	0.963	2	12	6	30.8
SVM (RBF)	0.993	0.995	0.969	0.982	1	6	49	10.4
SVM (linear)	0.994	0.995	0.974	0.984	1	5	38	2.9
Naive Bayes	0.968	0.886	0.954	0.919	24	9	3	1.0

Test set, spam = positive class. FP = legitimate messages flagged (of 845), FN = spam missed (of 195). Times are for fitting 4,160 messages and predicting 1,040.

	KNN	Naive Bayes	SVM
Strengths	• Simple, with no real training step; precision 0.989 with only 2 false alarms • A decision can be explained by showing similar past messages	• Fastest to train and predict (3 ms and 1.0 ms) • Recall of 0.954, above KNN, with almost no tuning	• Best on every metric: F1 0.982 (RBF) and 0.984 (linear) • Stable across folds (CV F1 0.986); the linear kernel needs 83 support vectors against 230
Weaknesses	• Lowest recall (0.938): 12 of 195 spam missed, 10 of them without a URL • Slowest to predict (30.8 ms vs 2.9 ms for linear SVM); k and scaling matter	• Lowest precision (0.886): 24 legitimate messages flagged, all containing a URL • Lowest F1 (0.919; cross-validated 0.927)	• Slowest to train (49 ms vs 3 ms for Naive Bayes) • RBF is hard to explain; both kernels need scaling and tuning (C, gamma)

2 Impact of probability assumptions in Naive Bayes

- **Independence does not hold.** Even inside one class, message length tracks URL count (r = 0.76 legit, 0.70 spam), URLs track digits (0.58, 0.69) and length tracks word count (0.38, 0.73).
- **Correlated evidence is counted several times.** A URL also lengthens a message and adds digits, so a legitimate message with a URL (about 8% of legit messages) gets an inflated spam score. All 24 false positives contain a URL.
- **Its confidence is unreliable.** Of 33 errors, 19 were made with over 90% confidence and 8 with over 99%.
- **Removing redundancy helps.** Dropping message_length and num_digits cut false positives from 24 to 10, lifted precision from 0.886 to 0.949 and halved log loss (0.111 to 0.052).



3 Trade-offs between interpretability and performance

Interpretability did not cost accuracy in this notebook: the linear SVM scored highest (accuracy 0.994 against 0.993 for RBF), and cross-validated F1 shows the two SVM kernels tied, well ahead of KNN and Naive Bayes:

Model	How a decision is explained	Cross-validated F1
SVM (linear)	One standardised weight per feature: more URLs (+2.6), more messages in 24 h (+2.6) and a newer account (−2.3 for account age) push toward spam	**0.986**
Naive Bayes	Each feature adds a likelihood ratio to the spam score, but correlated features are double counted	**0.927**
KNN	Shows the k most similar past messages; there is no global rule	**0.972**
SVM (RBF)	230 support vectors in a kernel space; weights cannot be read directly	**0.986**

Linear and RBF SVM tie on cross-validated F1 (0.986 and 0.986), but the linear kernel is the cleaner model: 83 support vectors against 230, and predictions about 3.5 times faster. Naive Bayes is easy to explain, but double counting distorts its explanation.

4 Business recommendation for real-world deployment

- **Deploy the linear SVM as the main filter.** Best test F1 (0.984), 1 false alarm among 845 legitimate test messages (0.1%) and 5 of 195 spam missed (2.6%). Its weights can be audited, and it predicts 1,040 messages in 2.9 ms.
- **Do not let Naive Bayes decide alone.** It would flag 2.8% of legitimate messages (24 of 845); use it, at most, as a cheap first-pass filter whose flags are reviewed.
- **Avoid KNN at scale.** It misses the most spam (6.2%) and its prediction cost grows with the stored training data.
- **Add content features back.** Missed spam is mostly URL-free (10 of KNN's 12 misses, 4 of the RBF SVM's 6), which behaviour alone catches poorly. Adding the two keyword scores cut test errors from 14 to 3 for KNN, 7 to 3 for RBF SVM, 6 to 2 for linear SVM and 33 to 5 for Naive Bayes. Check how num_special_chars is generated first: on its own it separates the classes perfectly, which suggests label leakage.
- **Operate with a cost-aware threshold.** Blocking a legitimate message usually costs more than missing some spam, so send borderline scores to human review. Monitor drift in sending behaviour, retrain regularly, and track the 7% of rows with missing sender data that were median-imputed.